

AEGIS Supplementary Material: Evaluation Metrics

TGLTW-RMIT

Purpose

This supplementary note defines every metric used in the AEGIS evaluation and ablation tables. Metrics are reported at the query, candidate, answer, or system level as indicated below.

Table 1: Evaluation metrics used by AEGIS.

Metric	Level	Definition and interpretation
Recall@k	Query / retrieval	Fraction of queries whose ground-truth item appears within the first k ranked candidates. We report Recall@1, Recall@5, and Recall@10 where available.
Mean Reciprocal Rank (MRR)	Query / retrieval	Mean of the reciprocal rank of the first relevant result: $MRR = N^{-1} \sum_i 1/r_i$. Higher is better; it rewards earlier correct candidates.
RRF score	Candidate / fusion	Weighted Reciprocal Rank Fusion score used to combine dense, lexical, and secondary-encoder rankings: $RRF(d) = \sum_m w_m / (k + \text{rank}_m(d))$.
VQA Exact Match	Answer / VQA	Percentage of questions for which the normalized generated answer exactly matches the reference answer.
VQA faithfulness	Answer / VQA	Percentage of answers supported by the resolved media evidence and the requested visual content.
Hallucination rate	Answer / VQA	Fraction of answers unsupported by valid media evidence or contradicted by the evidence. A fail-closed refusal is not counted as a hallucination.
Ambiguity score	Query / KIS-C	Combined ambiguity signal from Distinct Video Ratio (DVR) and Score Margin Ambiguity (SMA). Lower values indicate stronger convergence after clarification.
Ambiguity reduction	Query / KIS-C	Difference between the initial and final ambiguity scores: $A_{\text{initial}} - A_{\text{final}}$. Higher reduction indicates more effective conversational disambiguation.
Latency (p50/p95)	System / operation	Median or 95th-percentile wall-clock time for the complete measured operation. We report seconds for VLM workflows and milliseconds for retrieval paths.
Throughput (QPS)	System / concurrency	Completed query or candidate-scoring operations per second under the stated worker count.
Recall versus exact	Retrieval / precision ladder	Recall of an approximate HNSW mode relative to the exact brute-force ranking over the same collection.
Overall RAG score	Benchmark / corpus	Aggregate benchmark score combining retrieval quality, grounded answer quality, conversational behavior, and operational measurements under the declared evaluation protocol.

Reporting conventions

All percentages are reported on the evaluated query corpus and collection snapshot. Latency values include only the stages named in the corresponding experiment. Approximate HNSW results and exact brute-force results are compared using the same query vectors and candidate collection. For VQA, missing or invalid provenance produces the explicit fail-closed response UNKNOWN/N/A with zero confidence rather than an unsupported answer.